AI-Powered Customer Insight Engine for Beauty Brands

I. Introduction

As beauty brands like Fenty continue to release a wide range of makeup products, customer reviews play a vital role in understanding how well those products are received across different consumer groups. However, manually analyzing thousands of reviews to identify common issues, praise points, and sentiment trends is not scalable.

This project aims to build an AI-powered review analysis tool that enables beauty brands to extract structured insights from unstructured customer feedback. By combining sentiment analysis with topic modeling, we can uncover what customers are saying, how they feel, and why, helping teams make more informed decisions about product improvement, marketing, and consumer engagement.

II. General Overview of Role / Business

To address the challenge of analyzing large volumes of customer feedback, we developed an AI-driven review analysis framework using the following steps:

1. Sentiment Classification at Scale

What We Did: Used a RoBERTa language model to classify 1,000 customer reviews into positive, neutral, or negative categories.

How It Helps: Enables analysts to instantly understand overall customer sentiment without manually reading each review.

Why RoBERTa: RoBERTa is a modern language model that builds on BERT and offers stronger performance on real-world text. It's especially good at capturing the tone in short-form reviews, making it ideal for sentiment analysis in this context.

2. Review Cleaning and Consolidation

What We Did: Combined the pros and cons fields into a single opinion column and cleaned the text using standard NLP preprocessing techniques.

How It Helps: Creates a consistent input for modeling and ensures the output reflects meaningful consumer expression.

3. Uncovering Themes Using Topic Modeling

What We Did: Applied Non-negative Matrix Factorization (NMF) to identify key topics discussed in reviews.

How It Helps: Surfaces major discussion points such as texture, packaging, wearability, or shade match — offering deeper insight than sentiment alone.

Why NMF: We chose NMF because it's especially good at finding clear, meaningful topics in short, informal text like customer reviews, making it easier to interpret and act on the results.

4. Insight Generation Through Merging

What We Did: Merged the sentiment-labeled reviews with product metadata like brand and rating.

How It Helps: Allows product teams to compare customer sentiment with official ratings, detect mismatches, and prioritize underperforming SKUs.

III. Methods and Methodologies

Began by loading two datasets: one containing detailed customer reviews and another with makeup product metadata. The two were merged using a shared product_link_id key. For each review, we combined the pros and cons fields to form a complete opinion statement, which was then cleaned using regular expressions to remove punctuation and convert all text to lowercase.

Sampled 1,000 cleaned reviews and applied a RoBERTa language model to classify each as positive, neutral, or negative to perform sentiment analysis. These labeled sentiments were saved for further analysis.

Used topic modeling to uncover latent themes across customer feedback. This was done by vectorizing the cleaned text using a CountVectorizer and applying Non-negative Matrix Factorization (NMF) to extract the five most common topics customers discussed. This helped reveal not just whether customers liked a product, but why — highlighting key concerns and product features in an unsupervised manner.

Enriched dataset was prepared for business analysis by merging sentiment data back with product attributes such as brand, product name, and number of shades. This allowed us to investigate correlations between product performance and customer perception, laying the groundwork for actionable insights.

IV. Product Overview

Sentiment Classification Results

We applied the cardiffnlp/twitter-roberta-base-sentiment model to a sample of 1,000 customer reviews to categorize feedback into positive, neutral, or negative sentiment. This approach allowed us to quantify overall brand perception and detect emerging satisfaction trends across product lines. The sentiment labels were stored alongside each review and used throughout our analysis. This automated classification step eliminated the need for manual review while preserving the nuance of customer expression.

Topic Modeling Insights

To move beyond sentiment and uncover what specific product attributes customers were discussing, we used Non-negative Matrix Factorization (NMF) for topic modeling. By vectorizing cleaned review text and applying NMF, we extracted five key themes that frequently appeared in the dataset. These topics reflected core concerns such as shade match, hydration, wearability, coverage, and packaging. This unsupervised learning method gave us a deeper understanding of why customers felt the way they did, enabling product teams to focus on the most discussed features and potential areas for improvement.

Model Evaluation and Sentiment Trends

To assess the model's effectiveness, we evaluated the sentiment classification on the labeled sample of 1,000 reviews. The model achieved a strong accuracy of 93.8%, with an F1-score of 0.95 for positive sentiment and 0.93 for neutral sentiment. However, performance for negative sentiment was poor, with an F1-score of 0.00, largely due to class imbalance; only two reviews in the sample were negative. This limitation underscores the importance of more balanced sampling in future iterations to enhance model generalizability.

Using the sentiment-labeled dataset, we analyzed trends at the brand level to surface performance insights. IT Cosmetics, Benefit Cosmetics, and NUDESTIX consistently led in positive sentiment, suggesting strong brand loyalty and customer satisfaction. Brands like NARS and L'Oréal showed high volumes of neutral reviews, which could indicate less emotional engagement. Meanwhile, bareMinerals and BOBBI BROWN appeared frequently in the negative sentiment category, pointing to potential quality or communication issues. These patterns were further explored using monthly heatmaps and 3-month rolling averages, which revealed not only which brands sustained favorable perception but also those experiencing shifts over time. For example, NARS showed a sharp improvement in sentiment in 2025, while Lancôme and Laura Mercier revealed more volatile trends. These findings enable brands to tailor customer engagement strategies based on real-time feedback dynamics.

V. Technical Implementation

This project uses two datasets: one containing customer reviews and another with product-level metadata. These datasets were combined to analyze how customer sentiment relates to actual product features like ratings and inclusivity.

Customer Reviews (cleaned_makeup_reviews.csv):
This dataset contains 1,000 makeup product reviews that were processed using a RoBERTa model to generate sentiment labels (positive, neutral, negative). We used the truncated_review column to store cleaned text and the sentiment column to track model predictions. The product_name and brand columns were used to merge with product metadata.

Makeup Products (cleaned_makeup_products.csv):
This dataset includes structured information about makeup products. We used the rating column to analyze product satisfaction, and the num_shades column to assess inclusivity in foundation ranges. product_name and brand were cleaned and used to join this file with the review dataset.

To ensure a consistent merge, both datasets were cleaned by converting product and brand names to lowercase and stripping extra spaces. After merging, the unified dataset allowed for side-by-side analysis of sentiment and product attributes, enabling us to uncover insights into customer satisfaction and shade diversity.

VI. Bias Detection and Evaluation

While our model helps scale sentiment and topic analysis, it's important to recognize potential sources of bias in the data and modeling process. For example, certain brands or shade ranges may be overrepresented in the dataset, while others—especially those catering to underrepresented skin tones—may appear less frequently. This imbalance could lead the model to favor more common product types or overlook niche concerns.

Additionally, the language used in reviews can vary by demographic groups, which may affect how the sentiment model interprets tone or intent. If not monitored, this can skew the results and introduce unintended bias in insights generated.

To mitigate these risks, we recommend:

- Expanding the dataset to include a broader range of brands and shade offerings
- Monitoring sentiment trends across different product categories or shade families
- Flagging consistently low sentiment patterns for further investigation rather than assuming poor performance
- In future iterations, incorporating demographic or regional context to assess fairness more thoroughly

Addressing these factors helps ensure that the insights produced by the model support more inclusive, customer-centric decision-making.

VII. Future Implementation

There are several directions to expand this customer insight engine beyond its current capabilities. First, the sentiment classifier could be improved through fine-tuning with domain-specific makeup review data and expanding the labeled dataset, particularly for underrepresented sentiment classes. Additionally, the Streamlit interface could be deployed as a live web app with real-time API connectivity, enabling continuous ingestion of new reviews and sentiment updates.

We also see potential to integrate more advanced topic modeling techniques or large language models (LLMs) to provide richer insights, such as automatically summarizing customer pain points by product line. Further developments could include user segmentation, demographic-based analysis, and dynamic alerting when sentiment drops below a threshold for specific brands or product types. Over time, this tool could evolve into a full-scale voice-of-customer platform for real-time product feedback and customer experience optimization.

Vision for 5–10 Years

We envision this system evolving into a robust real-time customer insight engine embedded in CRM tools or e-commerce platforms. With ongoing integration of large language models (LLMs), it could provide automatic product diagnostics, monitor brand health continuously, and even recommend real-time customer engagement tactics. Beauty brands could use this to personalize offers, improve product lines faster, and reduce churn through proactive issue detection.

Adoption Benefits

Widespread adoption would:

- Automate high-volume review analysis for marketing/product teams.
- Provide early warnings on customer dissatisfaction.
- Surface underrepresented product issues more equitably.
- Shift companies from reactive to proactive customer experience strategies.

VIII. Challenges

On the technical side, working in Google Colab introduced environment limitations, particularly when testing Streamlit. The app required manual CSV uploads, and maintaining persistent UI access was difficult due to session timeouts and ngrok disconnections. These issues restricted the ability to deploy and iterate on the user interface in a more fluid, production-like setting.

Despite these obstacles, the core functionality was achieved: sentiment classification, topic detection, and visualization of insights through a working prototype dashboard.

IX. Conclusion

This project demonstrated how AI can transform unstructured customer reviews into structured, actionable insights for beauty brands. Using RoBERTa for sentiment classification and NMF for topic modeling, we uncovered not just how customers felt about products, but also why. Brands like IT Cosmetics and Benefit Cosmetics consistently earned high praise, while others such as bareMinerals showed signs of customer dissatisfaction.

While the model achieved strong performance for positive and neutral sentiment, it struggled with negative reviews due to class imbalance—highlighting the need for more balanced datasets in future work. Despite these limitations, our system successfully visualized sentiment trends over time, revealed brand performance shifts, and laid the groundwork for a scalable customer insight engine.

Ultimately, this AI-powered solution equips beauty brands with a deeper understanding of customer perception, helping them improve products, tailor marketing, and enhance overall customer satisfaction.